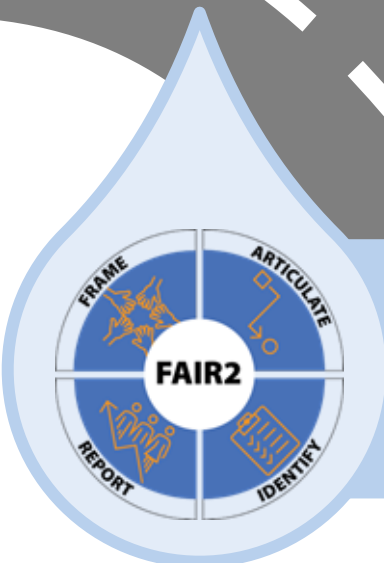


An illustration of Label Bias in Homelessness Data Analytics

Centering the Public Interest in Integrated Administrative Data Analytics

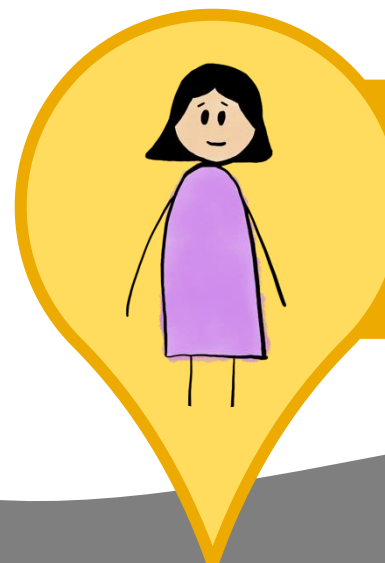
Module 3 of 5

Public Interest Technology University Network Year 6 Challenge
January 2026



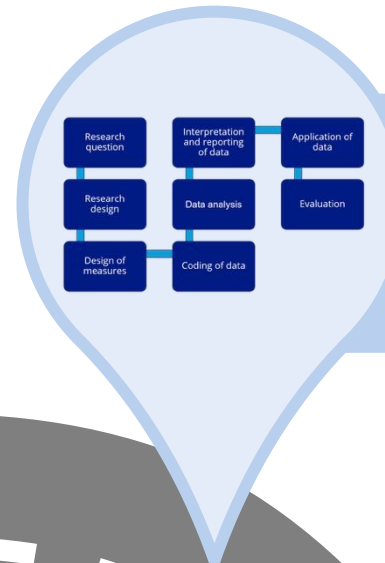
Module 1

FAIR2: A framework for integrating community knowledge in data analytics and addressing discrimination bias



Module 3

An illustration of Label Bias in Homelessness Data Analytics



Module 5

Recommendations to address constraints and bias in Integrated Data Analytics

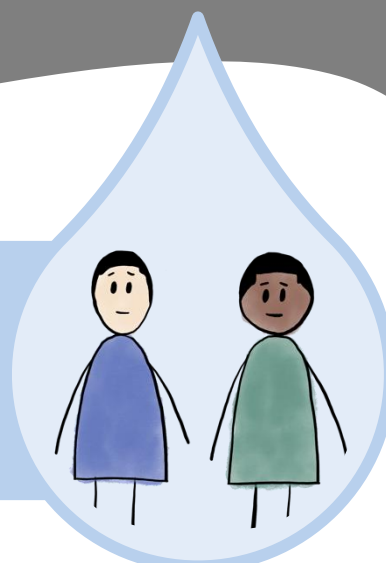
Module 2

Demographic Data in Integrated Data Systems: Seeking Experiential Knowledge to Inform Race and Gender Data



Module 4

An illustration of Collider Bias in housing research





Jesse has been on her own since her parents kicked her out at 17 years old, after she came out as a trans woman.



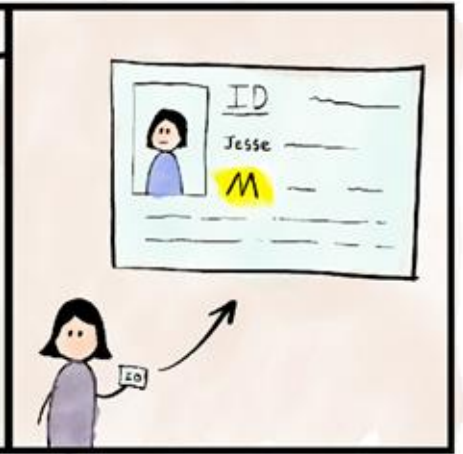
After staying with various friends, Jesse found herself in a relationship and moved in with her partner for several months.



However, the relationship suddenly went downhill, and it was not safe for Jesse to live with her ex anymore.



Jesse is not sure where to go and is running out of friends who will let her spend a night or two on their couch.



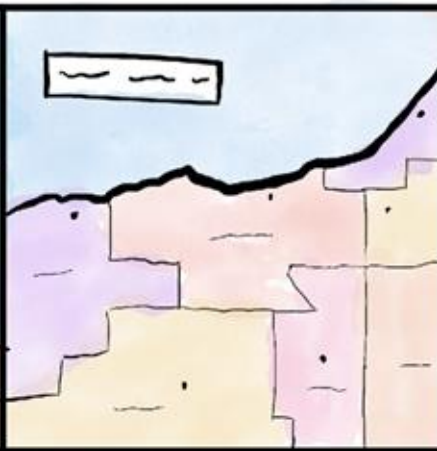
Jesse is looking into shelters but is scared to go as a trans woman. Plus, her ID still says "Male," and she doesn't feel safe in a men's shelter.



Casey is a social worker tasked with doing a needs assessment. They have a hunch that LGBTQ+ folks are in need of more services.



Casey pulls records from the biggest homeless shelter in the county for the past 5 years.



They plan to compare demographics of those serviced by the shelter to demographics in their county.



To Casey's surprise, very few LGBTQ+ folks and nearly zero trans folks were in shelter.



Casey concluded, "Wow! The LGBTQ+ community must be so strong that they find help within their community!"

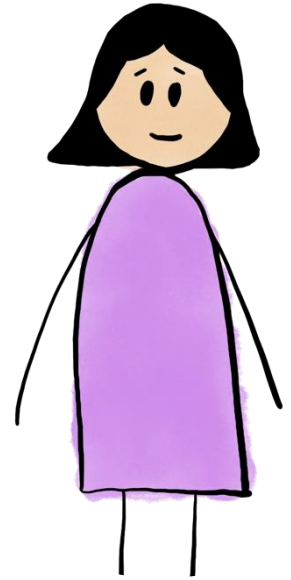
*Jesse shows us that there are multiple reasons why someone may not seek shelter, such as **safety** and **stigma**, which are particularly salient for marginalized populations.*

Think about other reasons why someone might not be able to seek shelter...

- *Safety*
- *Stigma*
- *Health issues*
- *(Dis)ability status*
- *Immigration status*

Remember what Casey concluded when looking only at shelter use data: “Wow! The LGBTQ+ community must be so strong that they find help within their community!”

What might Casey and other community workers be missing when looking at shelter data alone?



Introduction

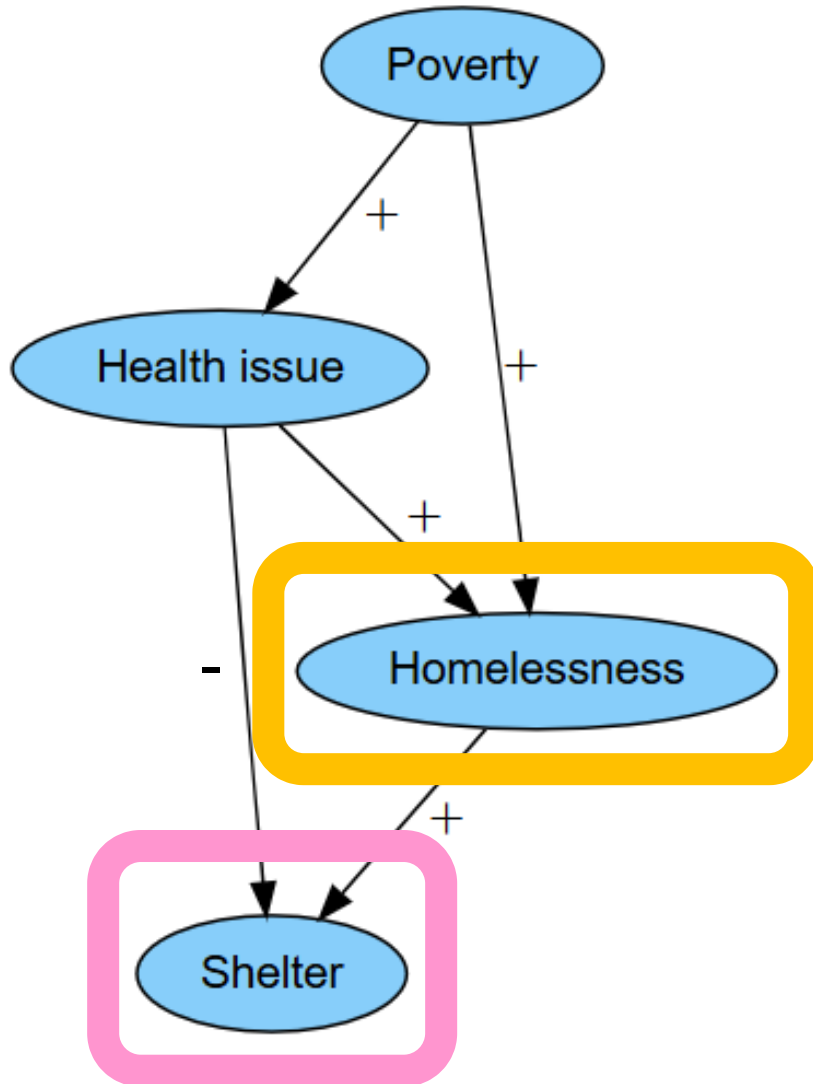
*You're a social worker who has worked with people who experienced **homelessness**, and oftentimes, you see people with chronic health issues and in poverty experiencing homelessness.*

Your team received administrative data on people who have used shelters in your county and is asked to analyze:

- 1. What are the **effects of poverty** on experiences of homelessness?*
- 2. What are the **effects of health status** on experiences of homelessness?*

You know that people may experience homelessness outside of shelter for multiple reasons and you want to alert the data analyst in your team that the administrative data may systematically miss more vulnerable people...

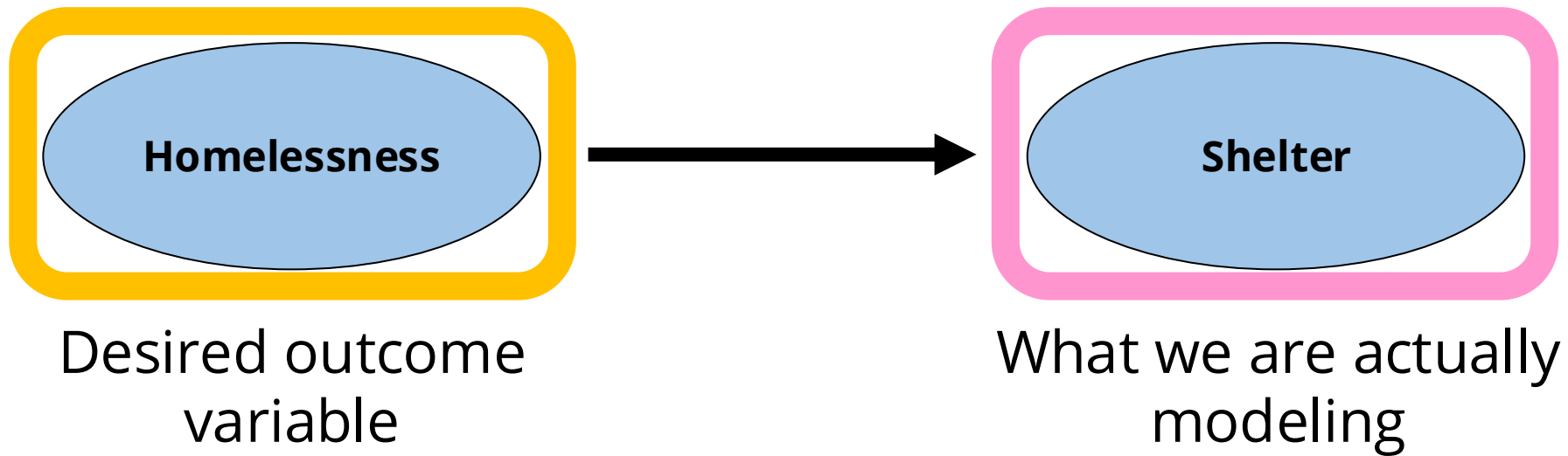
What is Label Bias



- We are interested on seeing how poverty and health relate to homelessness
- **Homelessness is our outcome variable of interest**, but our **measured variable (or label) is 'shelter use'**, an imperfect proxy for homelessness
- The graphic model represents the hypothesized relationships between our variables

What is Label Bias

Who gets left out?

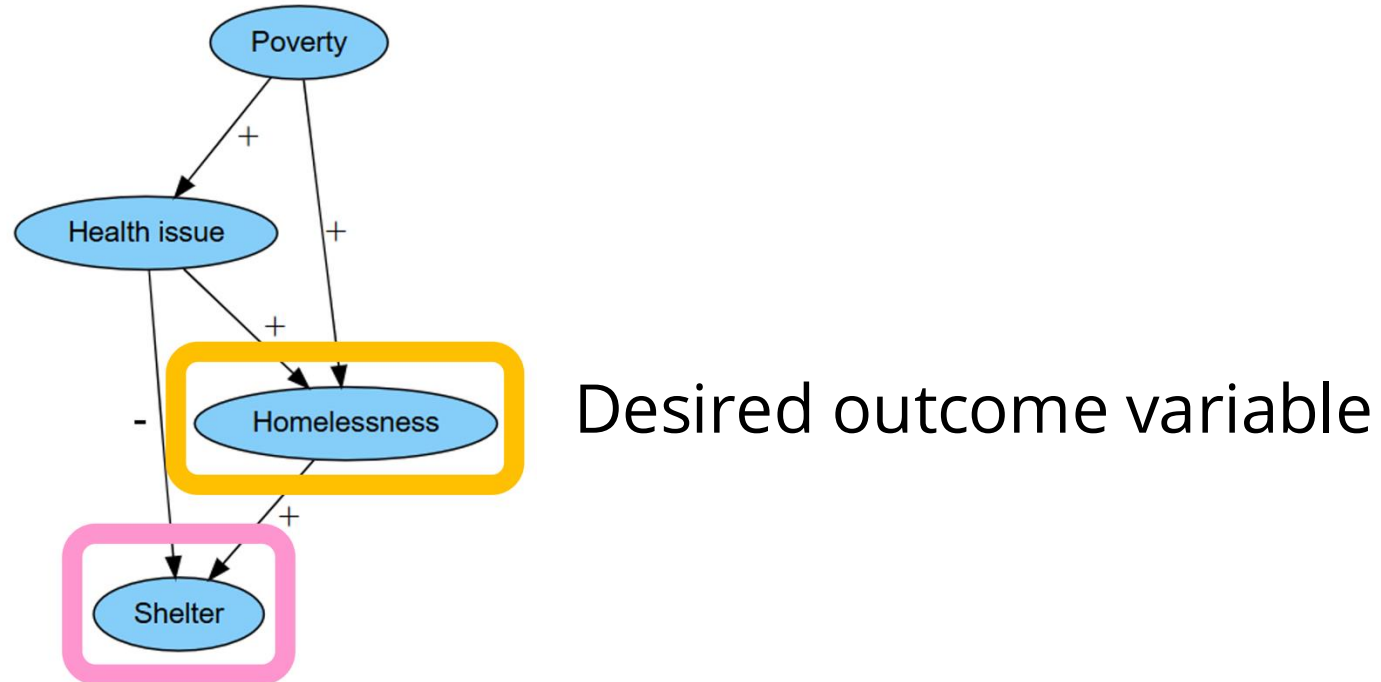


What is Label Bias

- Label bias occurs when the **variable used in data analytics** is only a **proxy label** for the **outcome of interest** that is difficult to measure (Mhasawade et al., 2024)
- Studying homelessness with data on shelter use, i.e. using **shelter use** as a proxy label of **homelessness** may lead to conclusions that:
 - Reproduce **historical inequities** and/or **human biases** (Banks et al., 2025)
 - Reflect **disparities** in society (Ceccon et al., 2025)

What is Label Bias

Who gets left out and why?
How does this reflect disparities/biases in society?

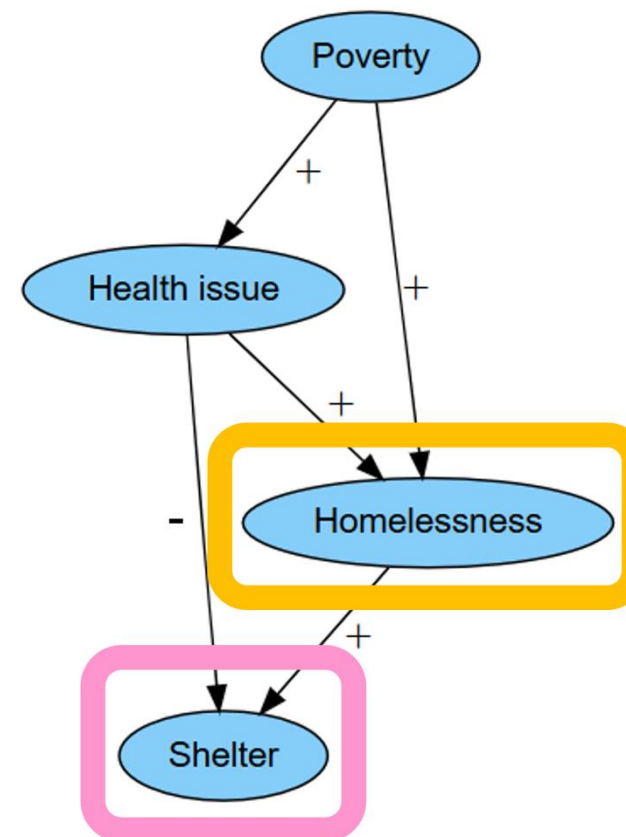


What we are actually measuring

Who is missing in the data?

"Your mental health comes first. If you're not there, then you're not going to actively seek help...your mental state should be taken care of because we can put people in houses or put them in shelters. But are they actively working with the psychiatrist? Are they actively working with the mental health [system]"

- Data Chat participant

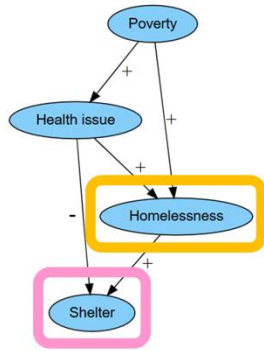


The potential negative relation between health issues and shelter is consistent with experiential knowledge about most vulnerable people being reluctant or unable to access homeless services.

Label Bias in Homelessness Research: Insights from Data Chat Participants

- Why would someone experiencing homelessness NOT seek shelter? (and subsequently not appear in our model)
 - **Safety**, especially if LGBTQ+ options are not available
 - **Stigma**
 - Jesse was struggling to decide if she should seek shelter because of both her safety as a trans woman and the social stigma that oftentimes accompanies marginalized gender identities
 - **Mental health**
 - Connection / support /Lack of relational component, feel like “just a number”
 - Services may not be **accessible**

“The worst thing is that you share your information with somebody and instead of them helping you, they’re judging and categorizing you.” - Data Chat participant

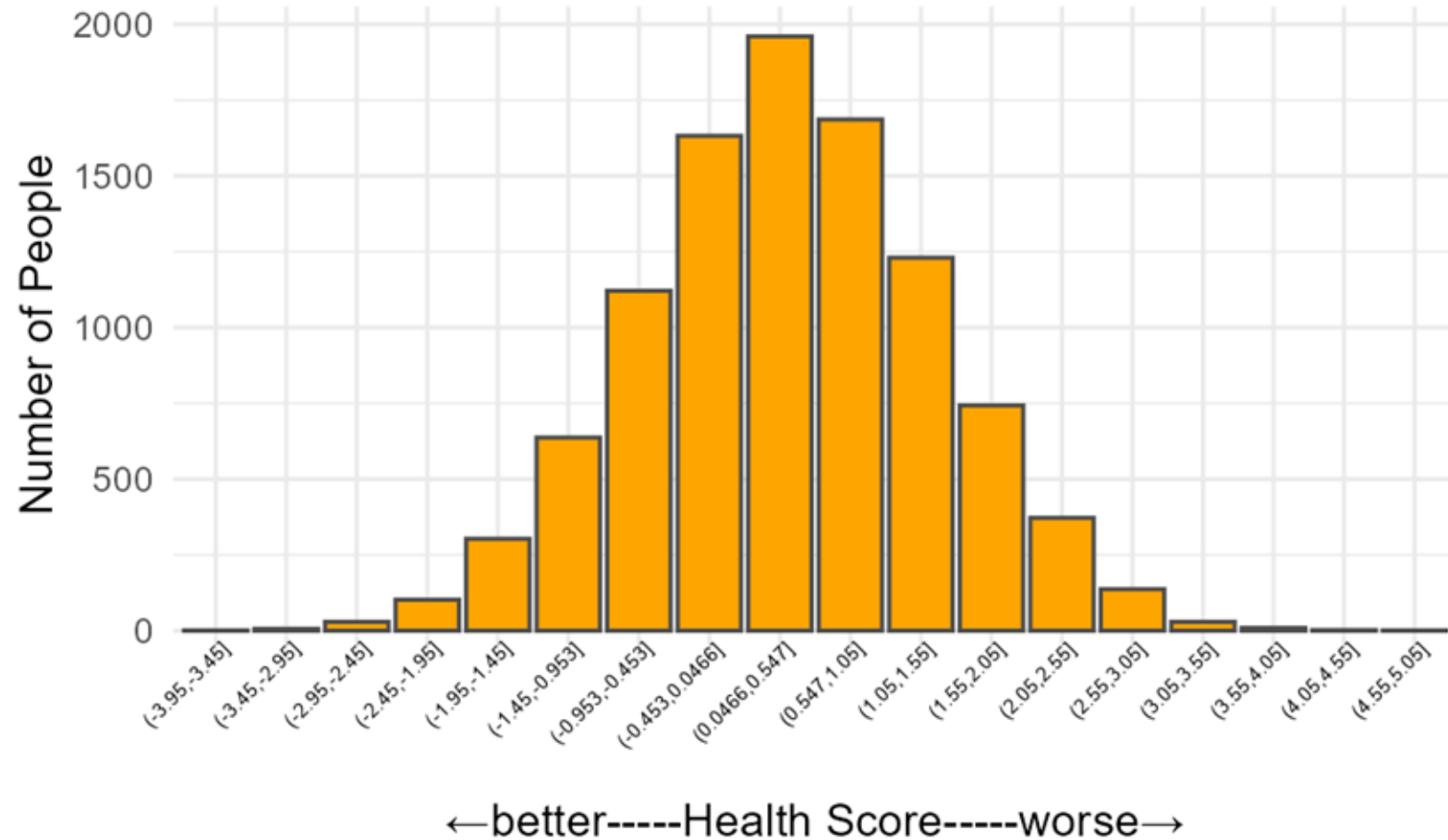
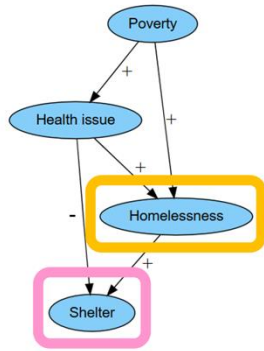


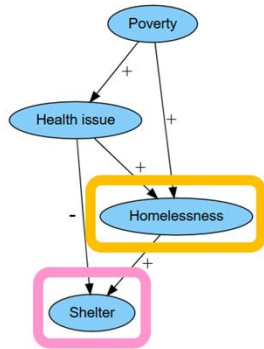
Simulated Data Example - I

- To see how label bias arises in the analysis, we **simulate individual-level data** on people's experience of poverty, health issues, homelessness, and shelter stay

Variable Name	Type of Variable	Meaning
Poverty	Binary	1 = Yes, 0 = No
Health Score	Continuous	Higher Values = Worse Health Score
Homelessness	Binary	1 = Yes, 0 = No
Shelter	Binary	1 = Yes, 0 = No

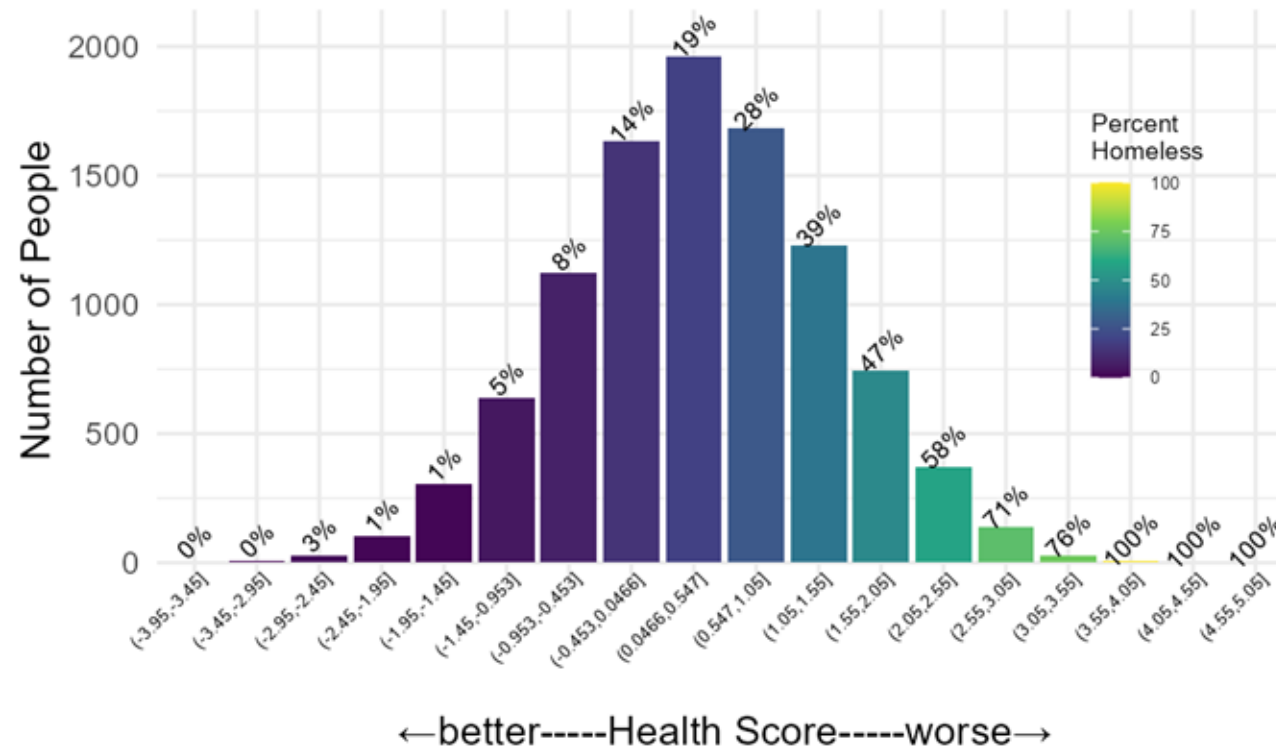
Simulated Data Example - I



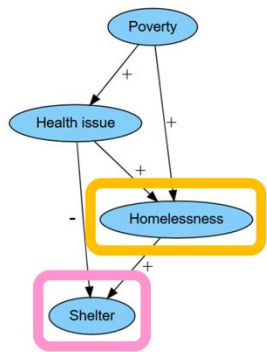


Simulated Data Example - I

- Then, we estimate the **effect of health issues on homelessness**



[Click here to see the simulation and results in R](#)



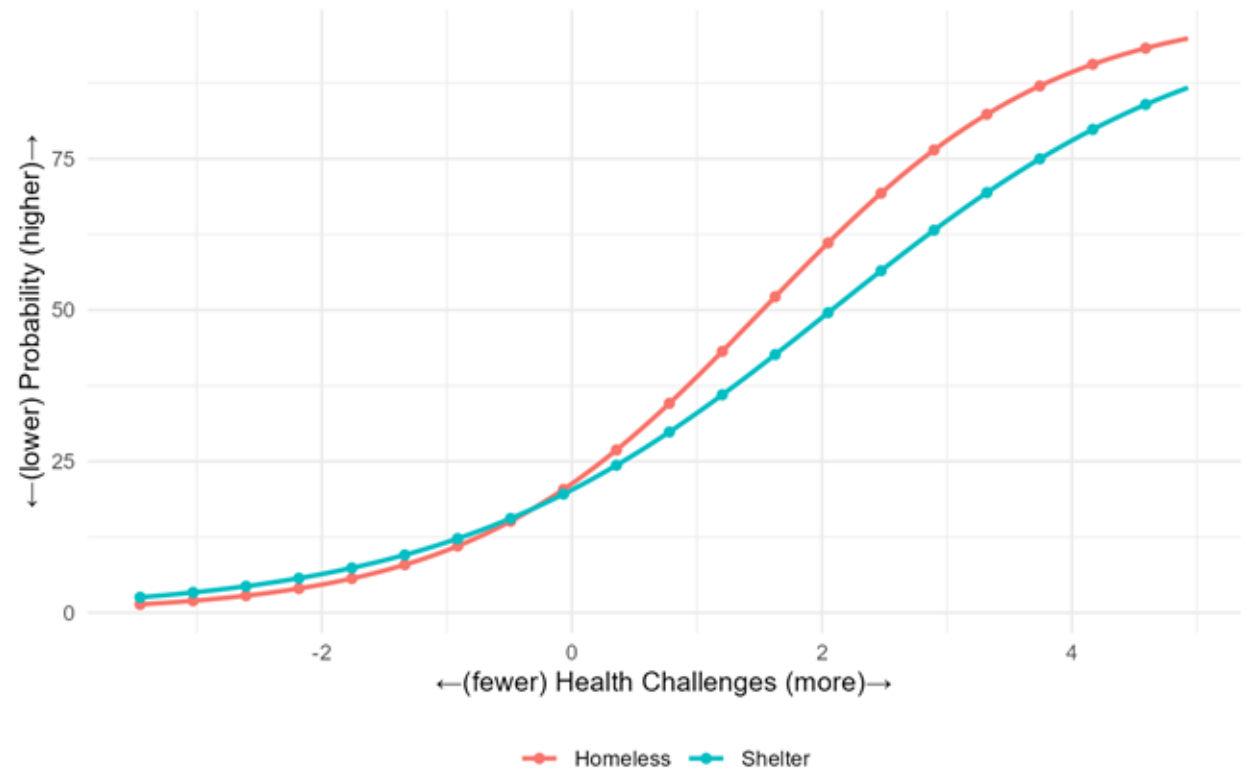
Simulated Data Example - II

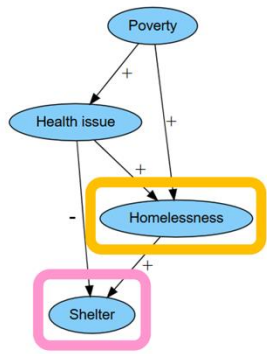
With this data, we run two **Logistic regression models** for those in poverty:

- **Model 1:**
 - Estimates the probability of Homelessness (y-axis) as a function of Health status (x-axis)
 - *See pink estimated curve*
 - This is the ideal model, but in reality, administrative records only include those in shelter
- **Model 2:**
 - Estimates the probability of Shelter use (y-axis) as a function of Health status (x-axis)
 - *See blue estimated curve*
 - This is a more realistic model, but it does not capture all individuals experiencing homelessness

The simulated data example shows that using shelter data as a **proxy** for homelessness will **underestimate** the risk that health issues pose for homelessness.

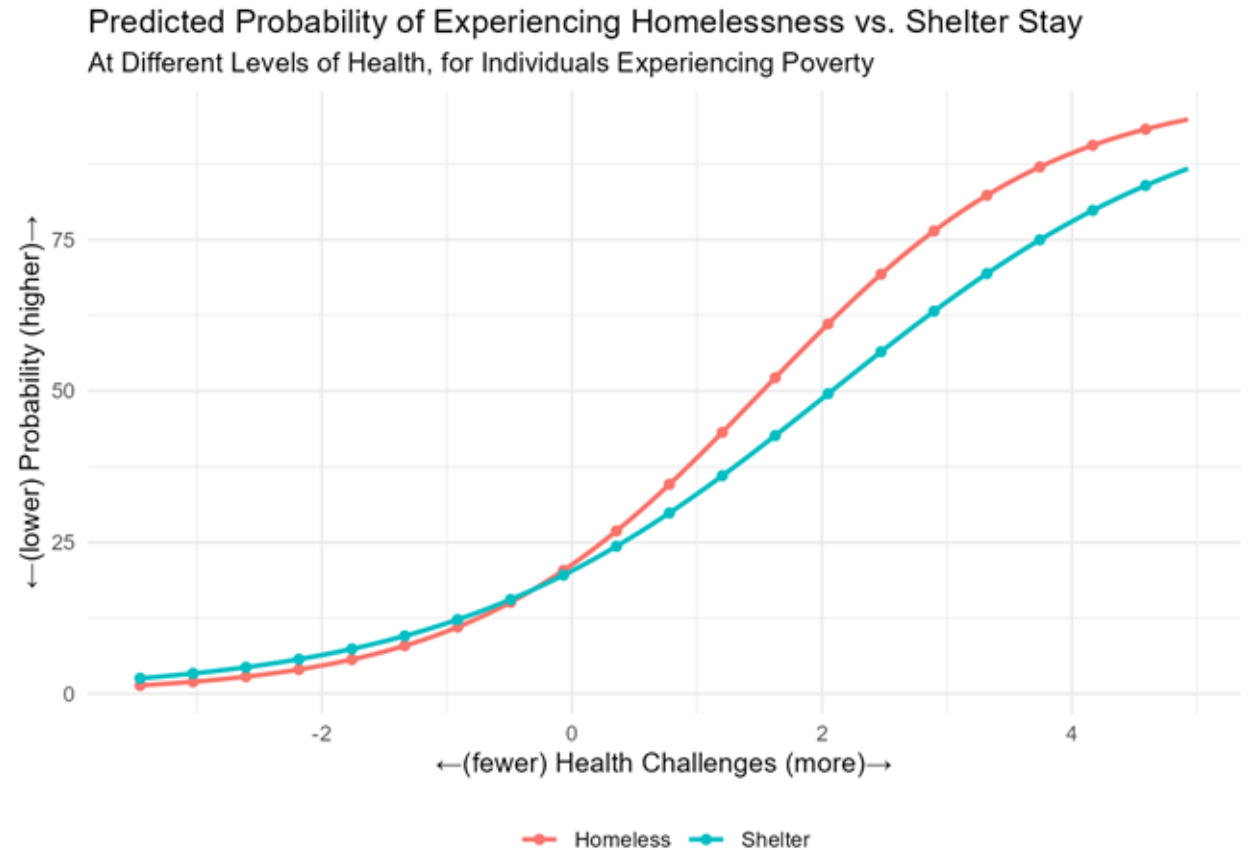
Predicted Probability of Experiencing Homelessness vs. Shelter Stay
At Different Levels of Health, for Individuals Experiencing Poverty





Simulated Data Example - II

- If people were prioritized for housing based on shelter data analytics, those with severe health conditions may rank lower for housing access than they would if the data captured all forms of homelessness, not just those visible through shelter use
- As a result, **the very individuals most vulnerable to poor outcomes could be deprioritized.**



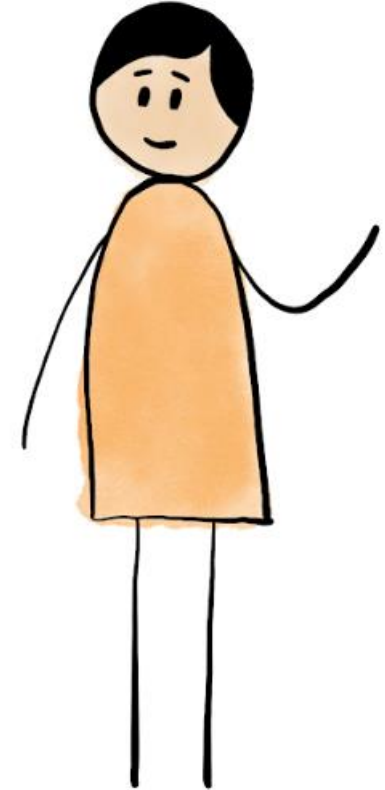
What to do about it?

- Use your social work experience to make data analytics aware of how data is **shaped by societal biases**
 - How was the data generated? What does data intake look like?
 - Contextualize data with **experiential knowledge** of those represented in the data
 - **Integrate health or other data** (i.e. food assistance data) where hidden homelessness can be identified
 - Use data and models to identify who is more likely to be left out of admin data or impacted by bias

"My personal feelings on it is that the only thing that needs to be identified is 'Yes, I'm human.'" - Data Chat participant

What to do about it?

- **Be transparent** about how data is shaped by societal biases and **seek experiential knowledge** to inform data analytics.
- **Casey's Realization:**
 - **I can't conclude that LGBTQ+ people don't need homeless services.**
 - I don't have data on all people experiencing homelessness, only on individuals who used shelters.
 - I need to understand the context around shelter use and who is left out of the data.



What to do about it?

- **Advocate for equitable & consistent data practices**
 - Inclusive options (ex. [SOGI question guidelines](#))
 - Accessibility
 - Provider training to limit implicit bias
 - Inform data practices with experiential knowledge (ex. [SOGI questions experiential knowledge](#))
- Advocate for **safer and more supportive housing services**, which will lead to a more equitable representation in administrative data, thus reduced label bias

"Being on the street, we're not thinking about the color we are, so why is the government thinking about what color we are?" - Data Chat participant

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